

HAT-P-27b

R-0044

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-27_260220 · created 2026-07-09T04:14:28+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461091.9087 +/- 0.0088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.04
Transit depth (Rp/Rs)^2	1.83 +/- 1.09 [%]
Orbital Inclination (inc)	87.91 +/- 2.9
Airmass coefficient 1 (a1)	27.51 +/- 15.6
Airmass coefficient 2 (a2)	-2.99 +/- 1.48
Scatter in the residuals of the lightcurve fit is	47.9 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	5.75
Transit Duration (day)	0.017 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.135 $\pm$ 0.04 vs 0.119 (deviation 0.4 σ)
Duration fit vs published	0.017 d vs 0.07 d (deviation 4.82 σ)
Mid-time O-C	-16.1 min
Depth SNR	3.4
Residual scatter	47.9 %

Light curve

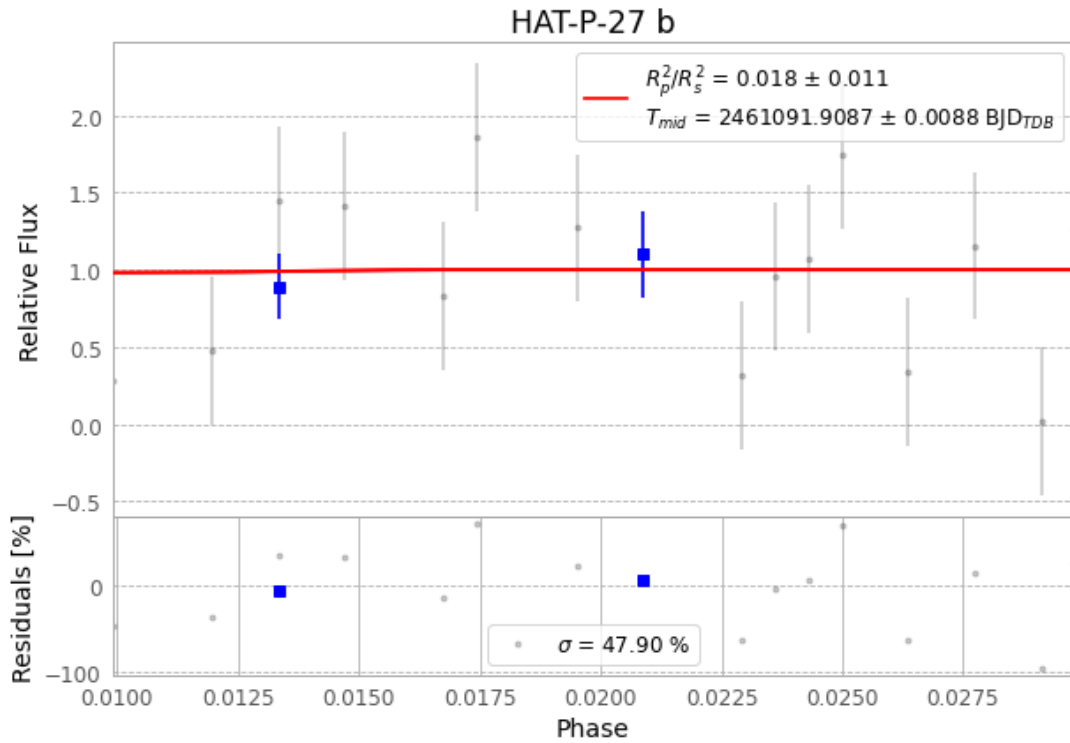


Figure 1 — FinalLightCurve_HAT-P-27 b_2026-02-20.png (SHA-256 b244d109090a53e6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [359, 169], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 74db3fc08dfe1072...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

31 FITS frames (20 MB), HATP-27260220101916.FITS → HATP-27260220115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-27-260220.log (2113adb08051...), FinalLightCurve_HAT-P-27 b_2026-02-20.png (b244d109090a...), FinalLightCurve_HAT-P-27 b_2026-02-20.csv (bcc915e625a8...)

Compute resources

Wall time 81.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.